
6 WATER QUALITY MODELLING

6.1 OVERVIEW

The water quality assessment for the Project was undertaken using the Model for Urban Stormwater Improvement Conceptualisation (MUSIC) developed by the Cooperative Research Centre for Catchment Hydrology (CRCCH). MUSIC estimates the pollutant loads and concentrations in stormwater runoff discharged from a representative catchment area. As the Project Area is within the Sydney Drinking Water Catchment, MUSIC was required to demonstrate that the Project will have a Neutral or Beneficial Effect (“NorBE”) on receiving environment water quality.

6.1.1 NorBE Criteria

MUSIC models the predicted changes in the total suspended solids (TSS), total nitrogen (TN) and total phosphorus (TP). Given the uncertainty in MUSIC modelling outcomes, WaterNSW requests a 10% improvement in post-development pollutant loads when compared with the pre-development pollutants (WaterNSW, 2023). A further criterion for pollutant loads requires that TSS, TN and TP concentrations must be equal to, or better, compared to the pre-development case pollutant concentrations for the 50th to 98th percentile cumulative frequency over the five-year modelling period.

6.2 MODEL CONFIGURATION

MUSIC modelling was undertaken for pre- and post-development conditions to estimate the pollutant loads discharged from the development footprint. The development footprint consists of the Project, which covers a total area of 74 ha and defines the catchment boundaries for modelling purposes. The development footprint has been broken into two (2) catchments as described in Table 6.1. Figure 6.1 illustrates the catchment locations for the development footprint.

It is noted that existing dams, gullies and some ridges are located outside the bounds of the development footprint. It is proposed that exclusion zones around these features be provided to ensure no disturbance to these areas. As the existing dams and ridges are to remain unchanged under post-development conditions, they have not been included within the catchment areas and have, therefore, been excluded from the MUSIC modelling. Additionally, the proposed treatment measures are to be offline and above or outside of the 2% AEP flood-affected areas as noted in Section 3.5.2 of “Using MUSIC in the Sydney Drinking Water Catchment” (WaterNSW, 2023). As the existing dams are generally located within the 2% AEP flood extent, exclusion from the modelling is further justified.

Table 6.2 outlines the amended water quality catchment areas which are to be included in the MUSIC modelling. Figure 6.2 illustrates the modelled catchment locations.

Table 6.1 Development Footprint Catchment Description

Catchment	Total Area (ha)	Description
2	25.10	Catchment 2 is the development footprint located to the west of the Wollondilly River and drains to the Wollondilly River. Under pre-development conditions this catchment consists of agricultural land (cropping and grazing). Under post-development conditions, this catchment consists of one (1) Construction Compound, a Substation, an Operation and Maintenance Facility (O&M Facility), a BESS area, Electrical Cabling and Access Tracks.
3	25.41	Catchment 3 is the development footprint located to the east of the Wollondilly River and drains to the Wollondilly River. Under pre-development conditions this catchment consists of agricultural land (cropping and grazing) and an existing unsealed Access Track. Under post-development conditions, this catchment consists of two (2) Laydown Areas, a Construction Compound, an administration area and an Access Track from the southern project boundary. The majority of the Access Track is an unsealed road.

Table 6.2 Modelled Water Quality Catchment Description

Catchment	Total Area (ha)	Description
2	25.10	Catchment 2 is located to the west of the Wollondilly River and drains to the Wollondilly River. Under pre-development conditions this catchment consists of agricultural land. Under post-development conditions, this modelled catchment consists of one (1) Construction Compound, a Substation, an Operation and Maintenance Facility (O&M Facility), a BESS area and new Access Tracks.
3	25.41	Catchment 3 is located to the east of the Wollondilly River and drains to the Wollondilly River. Under pre-development conditions, this catchment consists of agricultural land and an existing unsealed Access Track. Under post-development conditions, this catchment consists of two (2) Laydown Areas, a Construction Compound, an administration area and an unsealed Access Track from the southern project boundary. The majority of the Access Track is an existing unsealed road.

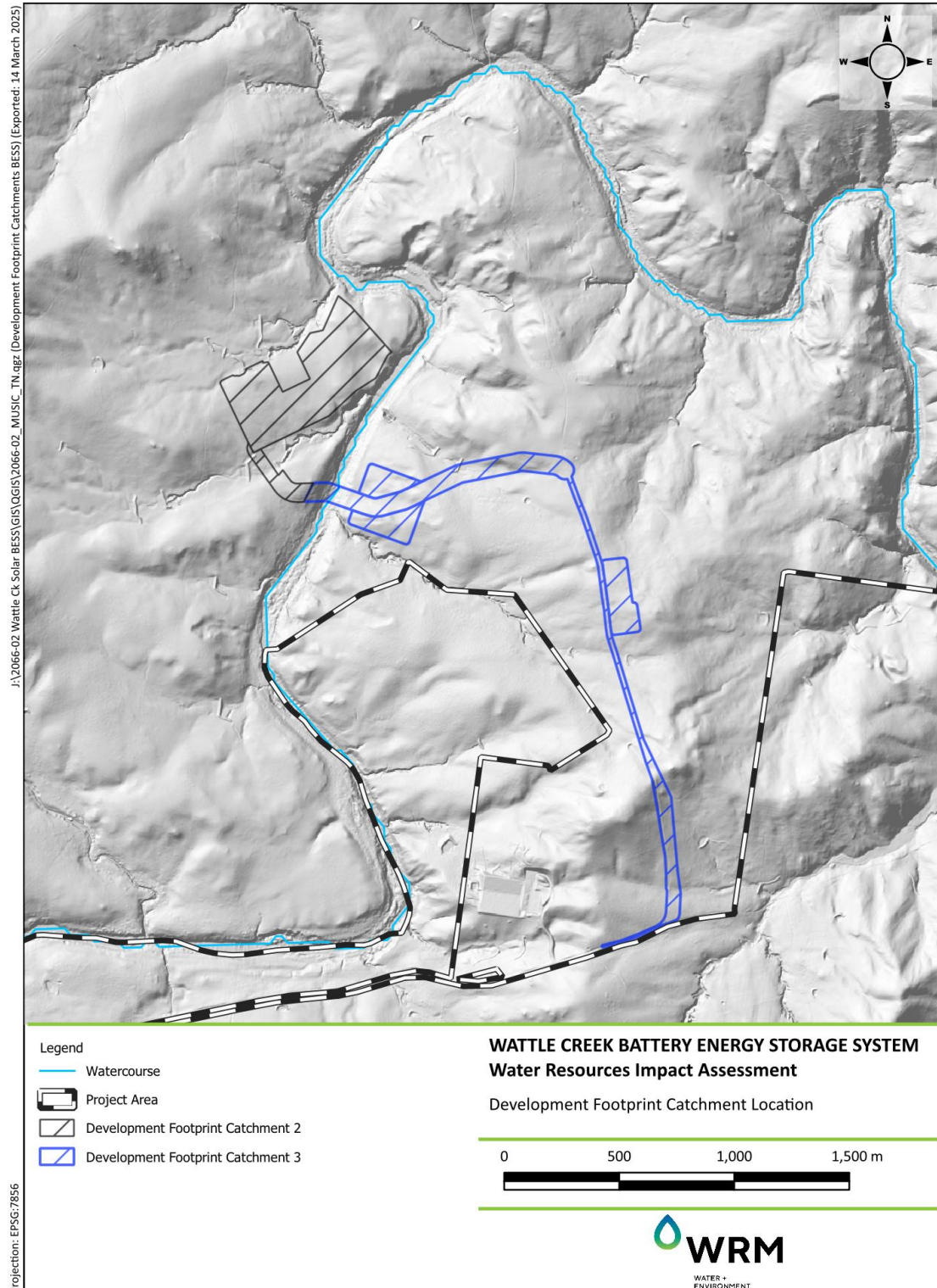


Figure 6.1 Development Footprint Catchment Locations

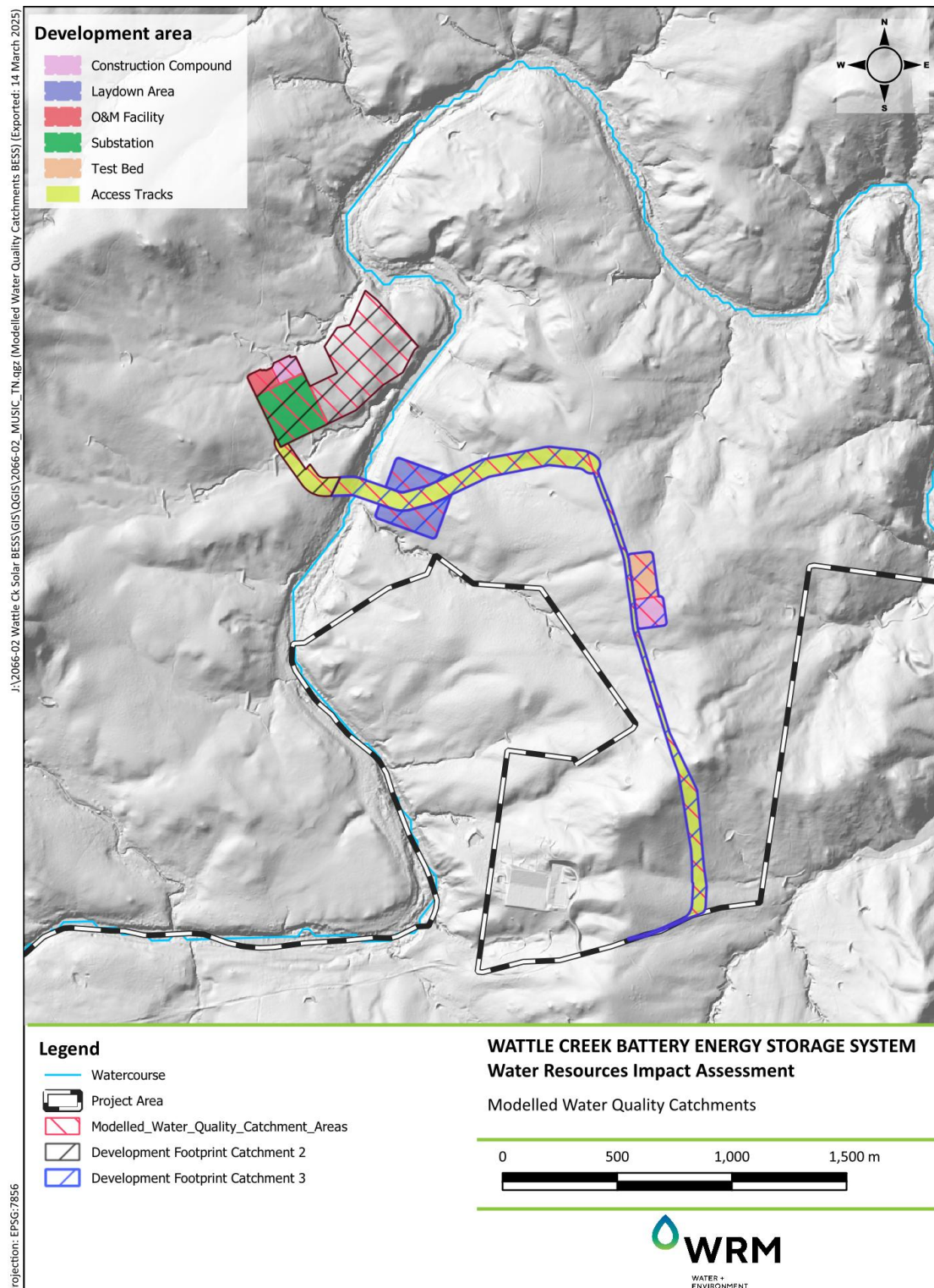


Figure 6.2 Modelled Water Quality Catchment Area for pre- and post-developed conditions

6.2.1 Meteorological Data

The Project Area is within the Wollondilly River catchment and is in Climate Zone 1 in Figure 3.1 of the *Using MUSIC in the Sydney Drinking Water Catchment* (WaterNSW, 2023). The meteorological MUSIC input data file data for Zone 1 was sourced from the WaterNSW website⁶. The provided data included average areal evapotranspiration estimates and six-minute rainfall depths. The supplied period covered 1 January 1995 to 31 December 1999. The provided data was loaded into MUSIC, as shown in Figure 6.3 and Figure 6.4

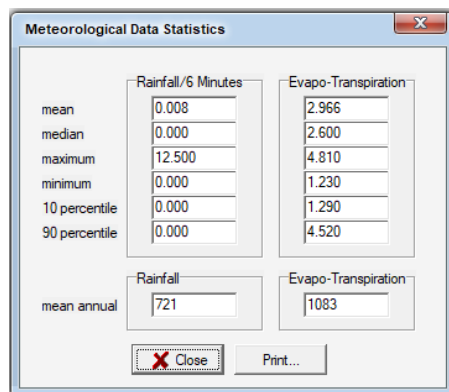


Figure 6.3 MUSIC Summary statistics of Meteorologic data

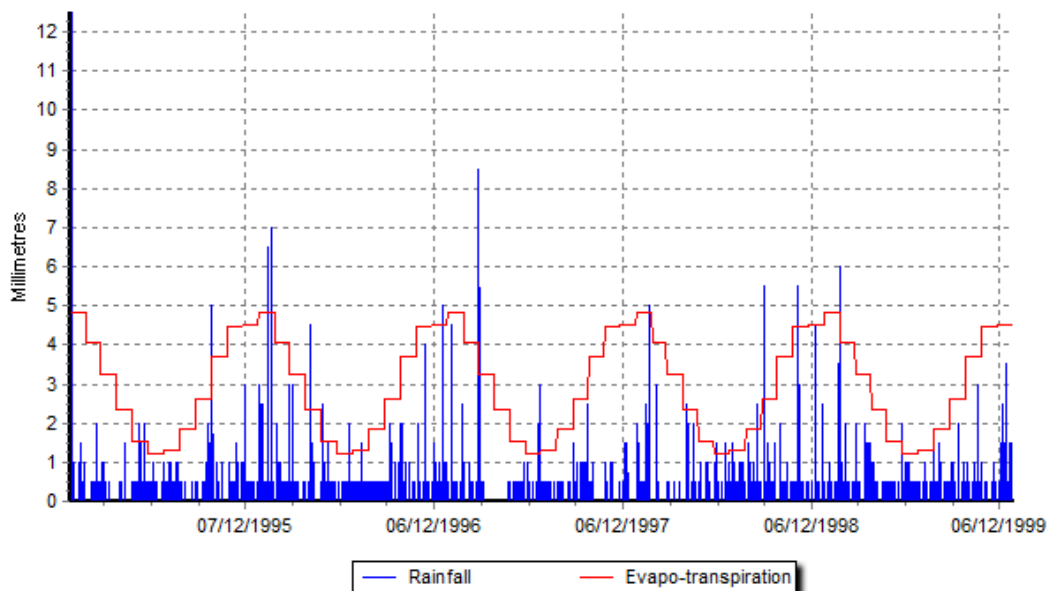


Figure 6.4 MUSIC Meteorologic time-series data

Source: WaterNSW

6.2.2 Model Configuration

The MUSIC model of the pre- and post-development was configured as shown in Figure 6.5 to enable the same stochastic seed to be utilised in both scenarios.

⁶ <https://www.watarnsw.com.au/water-services/catchment-protection/councils-and-developers>

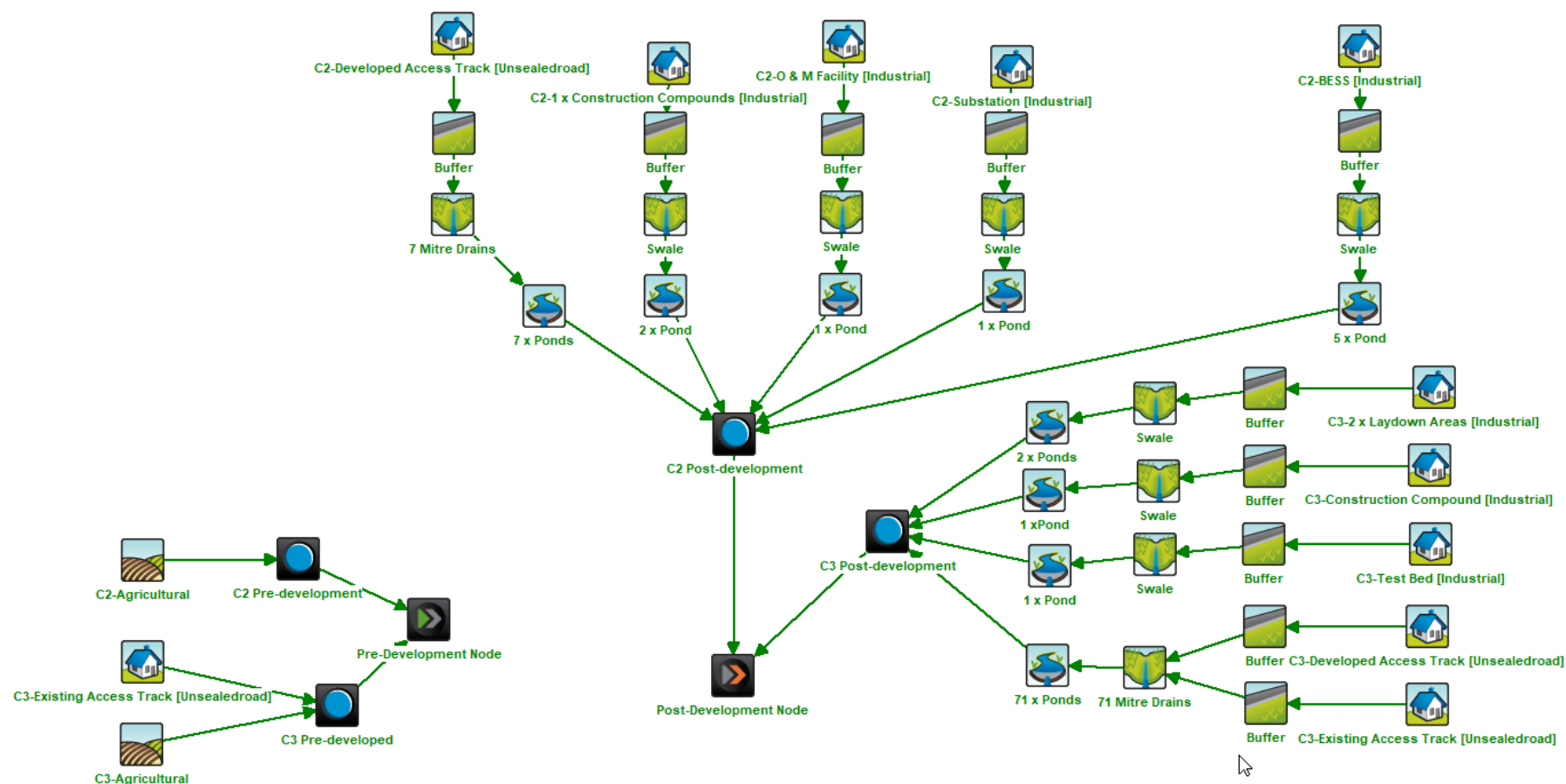


Figure 6.5 MUSIC Model Configuration

6.2.3 Source node parameters, areas and imperviousness

The modelled area is considered to be an agricultural land use, which is the source node in the water quality model. There are two soil types in the modelled area: Sodosols and Chromosols, which are based on the soil characteristic investigation by MineSoils. The soil types were identified in reference to the Australian Soil Classification (ASC) Scheme. The soil characteristics of the relevant sample location to the modelled area are presented in Table 6.3. Based on the survey results, the dominant soil types of the modelled area are sand, loamy sandy and clay.

Table 6.3 Soil type characteristics

Sample location	17	26	28	29	30
ASC Name	<i>Brown Sodosol</i>	<i>Brown Sodosol</i>	<i>Brown Chromosol</i>	<i>Brown Sodosol</i>	<i>Yellow Sodosol</i>
Permeability	Moderate	Moderate	Moderate	Moderate	Moderate
0.00 – 0.20	Loamy Sand	Loamy Sand	Loamy Sand	Loamy Sand	Loamy Sand
0.20 – 0.40	Loamy Sand	Medium Clay (0.20 – 0.55)	Sand (0.20 – 0.40)	Sand (0.20 – 0.30)	Loamy Sand (0.20 – 0.30)
0.40 – 0.65	Heavy Clay	Decomposing parent material 0.55+	Light Clay (0.40 – 0.70)	Heavy Clay 0.3+	Medium Clay (0.30 – 0.65)

Tables 4.4 and 4.5 in the WaterNSW 2023 document show soil and groundwater parameters, including soil storage capacity, field capacity, and runoff. Table 6.4 below shows the adopted parameter for this study according to the investigated soil data. It is noted that as there are three main soil types within a 0.5m root depth zone in the modelled area, Per WaterNSW 2023 Section 4.1.3.2, sand has been adopted for the Pervious area parameters as it will result in a lower estimate of flow volume for the pre-development condition.

Table 6.4 Soil and groundwater parameters

Rainfall-Runoff Parameters	Values
Pervious area properties	
Soil Storage Capacity (mm)	175
Initial Storage (% of Capacity)	25 (default)
Field Capacity (mm)	74
Infiltration Capacity Coefficient – a	360
Infiltration Capacity Coefficient – b	0.5
Groundwater properties	
Initial Depth (mm)	10 (default)
Daily Recharge Rate (%)	100
Daily Baseflow Rate (%)	50
Daily Deep Seepage Rate (%)	0

Source: WaterNSW, 2023

Table 4.6 and Table 4.7 in the WaterNSW (2023) document provide the defaults mean base flow and storm flow pollutant concentration and standard deviation for the source nodes in the MUSIC model. The default pollutant generation parameters presented in Table 6.5 were applied in this study.

Table 6.5 Adopted MUSIC pollutant concentration parameters

Land use type	Parameter	Total suspended solids (Log ₁₀ mg/L)		Total phosphorus (Log ₁₀ mg/L)		Total nitrogen (Log ₁₀ mg/L)	
		Base flow	Storm flow	Base flow	Storm flow	Base flow	Storm flow
Agriculture	Mean	1.30	2.15	-1.05	-0.22	0.04	0.48
	Std Deviation	0.13	0.31	0.13	0.30	0.13	0.26
Unsealed road	Mean	1.20	3.00	-0.85	-0.30	0.11	0.34
	Std Deviation	0.17	0.32	0.19	0.25	0.12	0.19
Industrial	Mean	1.20	2.15	-0.85	-0.60	0.11	0.30
	Std Deviation	0.17	0.32	0.19	0.25	0.12	0.19

Source: WaterNSW, 2023

6.3 PRE-DEVELOPMENT MODEL CONFIGURATION

MUSIC modelling of the pre-development scenario was undertaken to estimate the discharged pollutant loads and stormwater discharge concentrations from each pre-development MUSIC node. Table 6.6 presents the MUSIC source nodes corresponding to pre-development land features. It also presents the surface types, and associated parameters applied to the pre-development features.

Table 6.6 MUSIC source node parameter for pre-development

Catchment/MUSIC Node	Existing Feature	Source Node	MUSIC Surface Type	Area (ha)	Effective Impervious Area	Rainfall Threshold (mm)
C2- Agricultural Land	Agricultural Land	Agricultural	-	25.10	0%	1.0
C3- Existing Access Track	Existing Access Track	Urban	Unsealed road	7.16	50%	1.5
C3- Agricultural Land	Agricultural Land	Agricultural	-	18.25	0%	1.0

6.4 POST-DEVELOPMENT MODEL CONFIGURATION

MUSIC modelling of the post-development scenario was undertaken to estimate the discharged pollutant loads and stormwater discharge concentrations that would be produced from each MUSIC node, with the proposed Project infrastructure included.

Stormwater treatment measures were included in the modelled post-development scenario. Table 6.7 presents the MUSIC source nodes corresponding to each Project Design Feature under post-developed conditions. It also presents the surface types, and associated parameters applied to the post-development infrastructure surfaces.

Figure 6.2 illustrates the modelled water quality boundaries and land use mapping for the post-development scenario. Figure 6.5 shows the MUSIC model configuration for post-development conditions.

Table 6.7 MUSIC source node parameter for post-development

Catchment/MUSIC Node	Project Design Feature	Source Node	MUSIC Surface Type	Area (ha)	Effective Impervious Area	Rainfall Threshold (mm)
C2- Construction Compound	1 x Construction Compounds	Urban	Industrial	1.05	25%	1.5
C2-O&M Facility	O&M Facility	Urban	Industrial	1.02	25%	1.5
C2-Substation	Substation	Urban	Industrial	5.70	40%	1.5
C2-BESS	BESS	Urban	Industrial	15.14	40%	1.5
C2- New Access track	New Access Track	Urban	Unsealed road	2.19	35%	1.5
C3-Existing Access Track	Existing Access Track	Urban	Unsealed road	7.16	50%	1.5
C3-New Access track	New Access Track	Urban	Unsealed road	9.30	35%	1.5
C3-Laydown Areas	2 x Laydown Areas	Urban	Industrial	5.19	25%	1.5
C3-Construction Compound	1 x Construction Compound	Urban	Industrial	1.63	25%	1.5
C3-Administration area	Administration Area	Urban	Industrial	2.13	25%	1.5

6.4.1 Stormwater Treatment

The following stormwater treatment train is proposed for the development footprint under post-development conditions:

Existing and Proposed Access Tracks

- Vegetated buffer strips are proposed along the sides of the existing and proposed access tracks. Sheet flow runoff from the road will flow across the vegetated roadside strips removing coarse pollutants.
- Mitre drains are proposed at regular intervals along the proposed access tracks. Mitre drains will intercept concentrated flow from access track table drains and spread it out to level vegetated areas as sheet flow. This low velocity of sheet flow causes sediment to drop out. Mitre drains also provide a small amount of phosphorous reduction. A conservative estimate of a mitre drain every 50 m along existing and new access tracks is proposed. Each mitre drain is proposed to be 20m long with a base width of 0.5m, a top width of 1.5m, a depth of 0.3m and a vegetation height of 0.1m. The mitre drains have been modelled as a single treatment node with a total length calculated by multiplying the number of mitre drains by the proposed length of the mitre drain (No. of x 20m). Mitre drain spacing has been conservatively estimated based on Table 3-2 in Section 3.4.2 of the Water Sensitive Design Guide for Rural Residential Subdivision (WaterNSW, Feb 2023). Additionally, Access tracks should also be designed in general accordance with *ARRB Unsealed Roads Best Practice Guide Edition 2* (ARRB, 2020).

- Small ponds are to be located at the end of each mitre drain. The ponds will promote settling of medium to coarse grained particles. The retention time within the pond ensures treatment of nutrients. Each pond is proposed to have a 2 m² surface area, a 3 m³ pool volume, estimated 1 m³ initial volume, and average depth of 1 m and a notional detention time of greater than 48 hours. If more than 1 pond is proposed the above parameters are multiplied by the number of proposed ponds.
- Table 6.8 outlines further details of this treatment train for the corresponding modelled nodes.

Construction Compounds, O & M Facility, Laydown Areas, Substation, BESS and Administration Area

- Vegetated buffer strips are proposed along the perimeter of the development footprint. Sheet flow runoff will flow across the vegetated strips removing coarse pollutants.
- Vegetated swale drains are to be located adjacent to buffer strips along the downstream side of the development footprint. The vegetated swales will convey and filter stormwater runoff through vegetation to remove coarse sediment and total suspended solids. Each vegetated swale drain is proposed to have a base width of 1 m, a top width of 5 m, a depth of 0.35 m and a vegetation height of 0.25 m. The length of each vegetated swale drain will vary. Refer to Figure 6.6 for typical swale layout.
- Ponds are to be located at low points at the end of the proposed vegetated swales. The ponds will promote settling of medium to coarse grained particles. The retention time within the pond ensures treatment of nutrients. Each pond is proposed to have a 78.5 m² surface area, a 170 m³ pool volume, estimated 85 m³ initial volume, and average depth of 1.5 m and a notional detention time of greater than 48 hours. If more than 1 pond is proposed the above parameters are multiplied by the number of proposed ponds. Refer to Figure 6.6 and Figure 6.7 for typical pond layout.
- Table 6.8 outlines further details of this treatment train for the corresponding modelled nodes.

Table 6.8 Stormwater Quality Treatment Measures for Post-development Conditions

Catchment/MUSIC Node	Treatment Train	Description/Parameters
C2 – One (1) x Construction Compound	Buffer → Swale → Pond	Swale drain with: ○ Total length of 412 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m
		<u>2 Ponds with:</u> ○ Total surface area of 157 m² ○ Each pond has an extended detention depth of 2m ○ Total Permanent Pool Volume of 340 m³ ○ Initial Volume of 170 m³

Catchment/MUSIC Node	Treatment Train	Description/Parameters
C2 – O & M Facility	Buffer → Swale → Pond	Swale drain with: ○ Total length of 108 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m <u>1 Pond with:</u> ○ Total surface area of 78.5 m² ○ Extended detention depth of 2m ○ Total Permanent Pool Volume of 170 m³ ○ Initial Volume of 85 m³
C2 - Substation	Buffer → Swale → Pond	Swale drain with: ○ Total length of 320 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m <u>1 Pond with:</u> ○ Total surface area of 78.5 m² ○ Extended detention depth of 2m ○ Total Permanent Pool Volume of 170 m³ ○ Initial Volume of 85 m³
C2 - BESS	Buffer → Swale → Pond	Swale drain with: ○ Total length of 1243 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m <u>5 Ponds with:</u> ○ Total surface area of 392.5 m² ○ Each pond has an extended detention depth of 2m ○ Total Permanent Pool Volume of 850 m³ ○ Initial Volume of 425 m³
C2 - New Access track	Buffer → Mitre Drains → Pond	<u>7 Mitre Drains with:</u> ○ Total length of 140 m ○ Equates to 1 Mitre Drain every 50 m ○ Base with 0.5 m; Top width 1.5 m; Depth of 0.3 m; Vegetation height of 0.1 m <u>7 Ponds with:</u> ○ Total surface area of 14 m² ○ Each pond has an extended detention depth of 2m ○ Total Permanent Pool Volume of 21 m³ ○ Initial Volume of 7 m³

Catchment/MUSIC Node	Treatment Train	Description/Parameters
C3 - Existing Access Track & C3 - New Access track	Buffer → Mitre Drains → Pond	<u>71 Mitre Drains with:</u> ○ Total Drain length of 1,420 m ○ Equates to 1 Mitre Drain every 50 m ○ Base with 0.5 m; Top width 1.5 m; Depth of 0.3 m; Vegetation height of 0.1 m
		<u>71 Ponds with:</u> ○ Total surface area of 142 m² ○ Each pond has an extended detention depth of 2m ○ Total Permanent Pool Volume of 213 m³ ○ Initial Volume of 71 m³
C3 – Two (2) x Laydown Areas	Buffer → Swale → Pond	Swale drain with: ○ Total length of 863 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m
		<u>2 Ponds with:</u> ○ Total surface area of 157 m² ○ Each pond has an extended detention depth of 2m ○ Total Permanent Pool Volume of 340 m³ ○ Initial Volume of 170 m³
C3 - Construction Compound	Buffer → Swale → Pond	Swale drain with: ○ Total length of 357 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m
		<u>1 Pond with:</u> ○ Total surface area of 78.5 m² ○ Extended detention depth of 2m ○ Total Permanent Pool Volume of 170 m³ ○ Initial Volume of 85 m³
C3 – Administration Area	Buffer → Swale → Pond	Swale drain with: ○ Total length of 317 m ○ Base with 1 m; Top width 5 m; Depth of 0.5 m; Vegetation height of 0.25 m
		<u>1 Pond with:</u> ○ Total surface area of 78.5 m² ○ Extended detention depth of 2m ○ Total Permanent Pool Volume of 170 m³ ○ Initial Volume of 85 m³

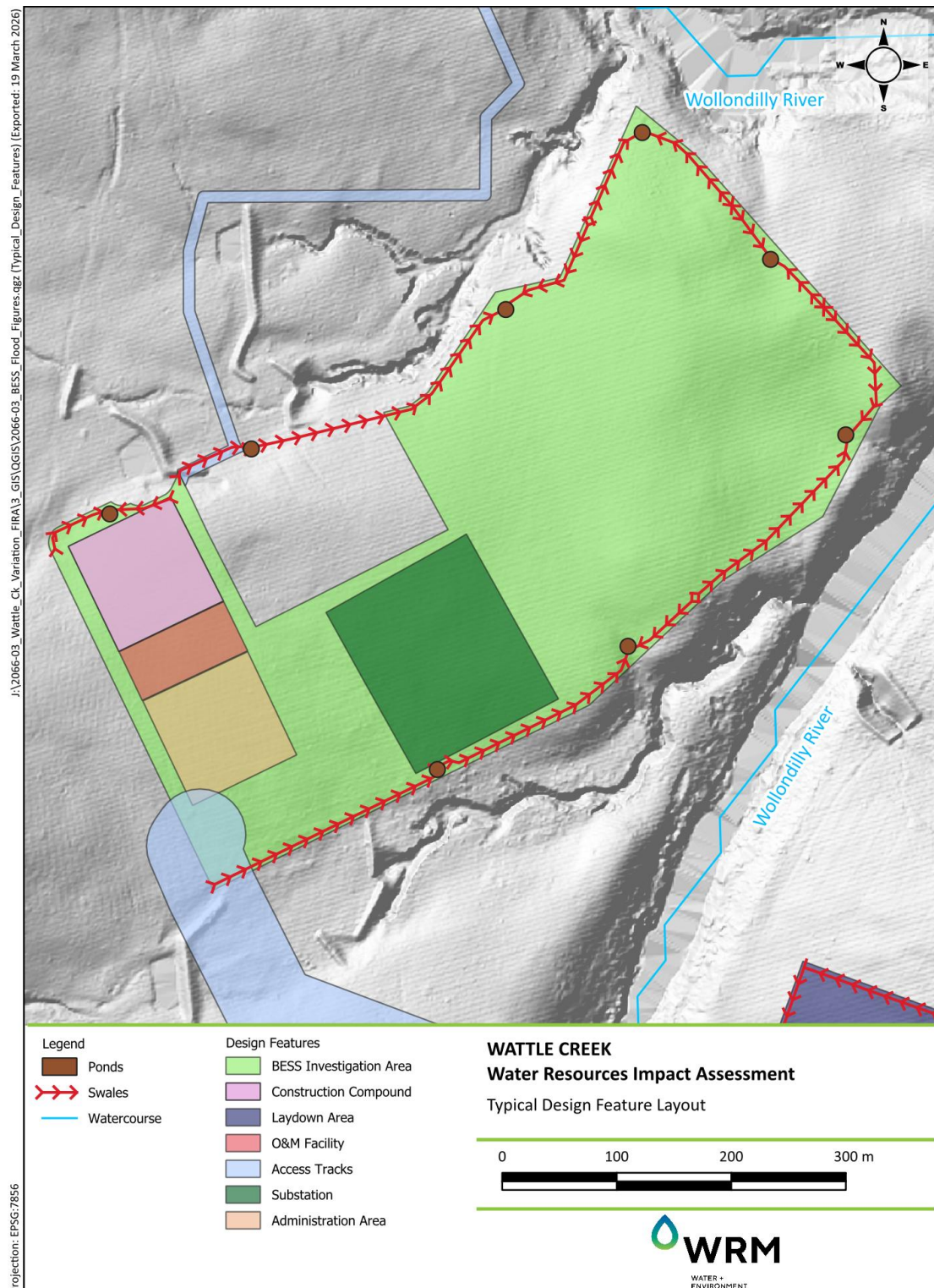


Figure 6.6 Typical Design Feature Swale and Pond layout 1

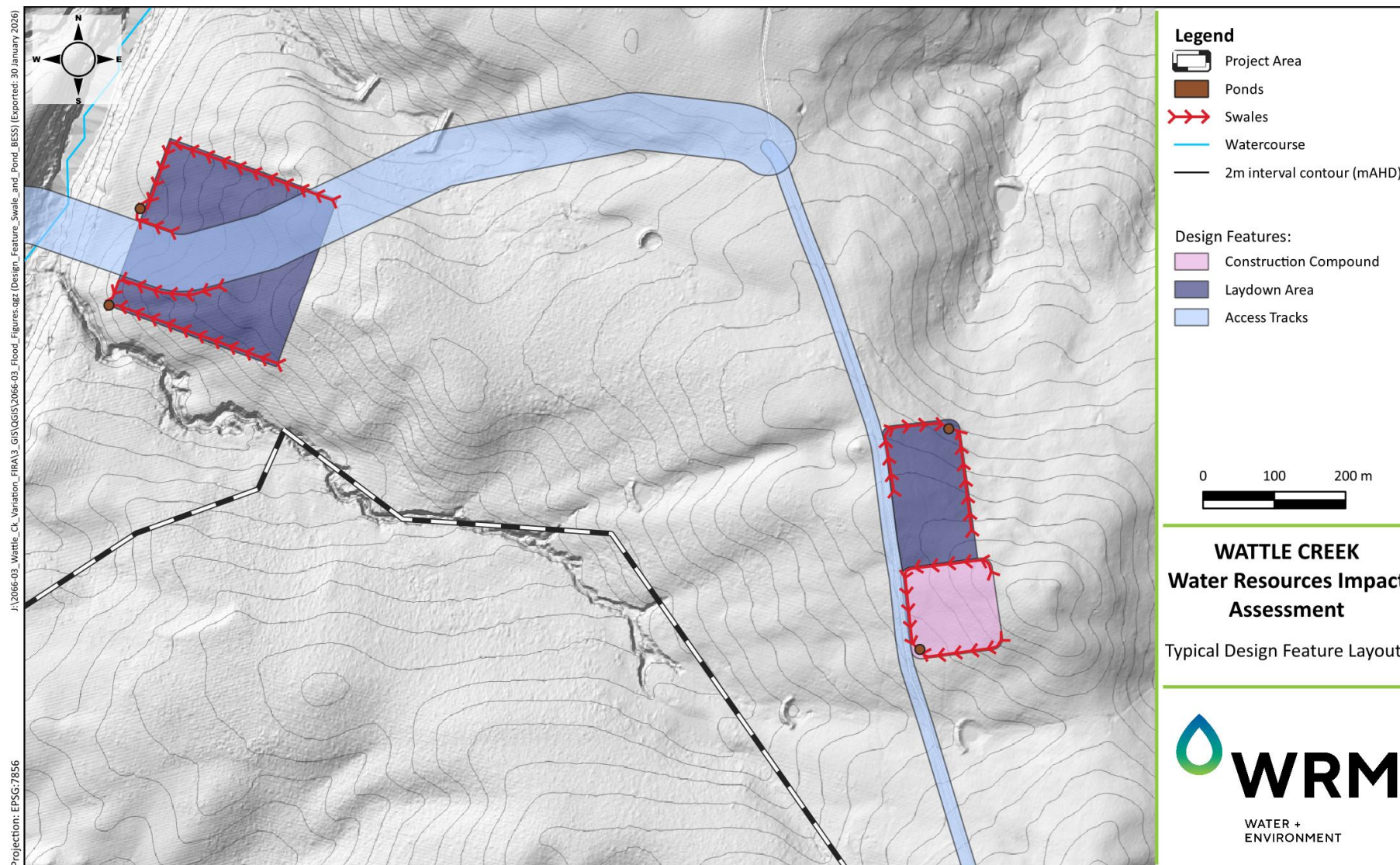


Figure 6.7 Typical Design Feature Swale and Pond layout 2

6.5 WATER QUALITY MODELLING RESULTS

6.5.1 Pollutant Loads

Table 6.9 presents a comparison of the modelled mean annual pollutant loads for the pre- and post-development (with treatment) scenarios, including the estimated percentage reduction in pollutant loads. Modelling predicts reductions in annual pollutant loads for TSS and Gross Pollutants. These reductions surpass the NorBE requirement for a 10% beneficial impact and are therefore NorBE compliant.

While a reduction in TP is predicted, the reduction does not meet the NorBE requirements. The MUSIC modelling of the post-development scenario also expects there would be an increase in annual pollutant loads for TN. A reduction in TN and TP could be achieved through the construction of tertiary treatment measures - such as bio-retention basins or constructed wetlands. Table 6.1 within WaterNSW (2023) specifies the suitability of treatment measures. Bio-retention basins are indicated to be unsuitable due to the scale of this Project. For the scale of this Project constructed wetlands are suitable, however the use of such measures is not feasible given that the Project Area does not possess large and level expanses for their construction. Additionally, Section 4.5.2 (WaterNSW, 2023) requires a constructed wetland to be located outside of the 2% AEP flood envelope.

At present, the land usage of the Project Area comprises cropping and grazing. Both activities would involve the application of fertilisers and the biological fixation of nitrogen. Within the post-development scenario, sources of nitrogen would remain unchanged or would be lower due to the construction of infrastructure. Therefore, TN predicted by the MUSIC model's parameter set appears to be demonstrating an overly low pre-development quantity, and possibly an overestimated TN load for post-development conditions. This anomaly should be investigated further prior to the implementation of tertiary treatment trains.

Should further treatment train measures be required to satisfy NorBE criteria for TN and TP, it is recommended that Project stormwater treatment measures be considered during the detailed design phase with a refined MUSIC model that reflects the detailed Project design.

Table 6.9 WQ Study Area Comparison: Mean Annual Pollutant Loads

Pollutant	Pre-development (kg/year)	Post-development (kg/year)	Difference	Load Reduction	Neutral or beneficial effect
TSS	27,300	2,810	-24,490	90%	True
TP	17.1	16.5	-0.60	4%	False
TN	109	196	87	-	False
Gross Pollutants	847	0	-847	100%	True

6.5.2 Pollutant Concentrations

Figure 6.8, Figure 6.9, and Figure 6.10 present comparisons of the predicted TSS, TP and TN concentrations, respectively, for the pre- and post-development (with treatment) scenarios and indicate:

- TSS concentrations are equal or better for the post-development scenario between the 5th and 98th percentile. Post-development TSS concentrations achieve the NorBE criteria.

- Post-development TP concentrations do not achieve the NorBE criteria of being better than the pre-development scenario between the 50th and 98th percentiles. TP concentrations are better for the post-development scenario above the 60th percentile result.
- Post-development TN concentrations do not achieve the NorBE criteria of being better than the pre-development scenario between the 50th and 98th percentiles. TN concentrations are better for the post-development scenario above the 80th percentile result.

As such, additional treatment train measures may be required to satisfy NorBE criteria for TN. Additional Project stormwater treatment train measures should be considered during the Project's detailed design phase with a refined MUSIC model that reflects the detailed Project design.

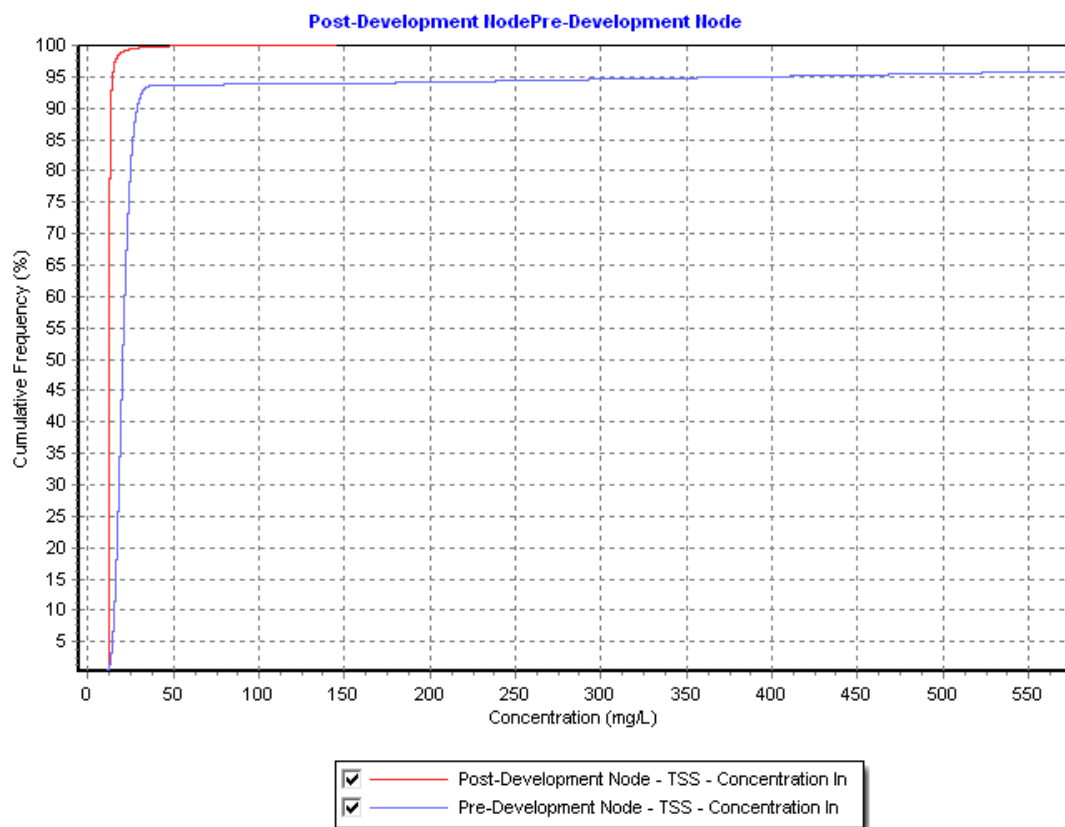


Figure 6.8 Pre-development and Post-development Comparison: Predicted TSS Concentrations

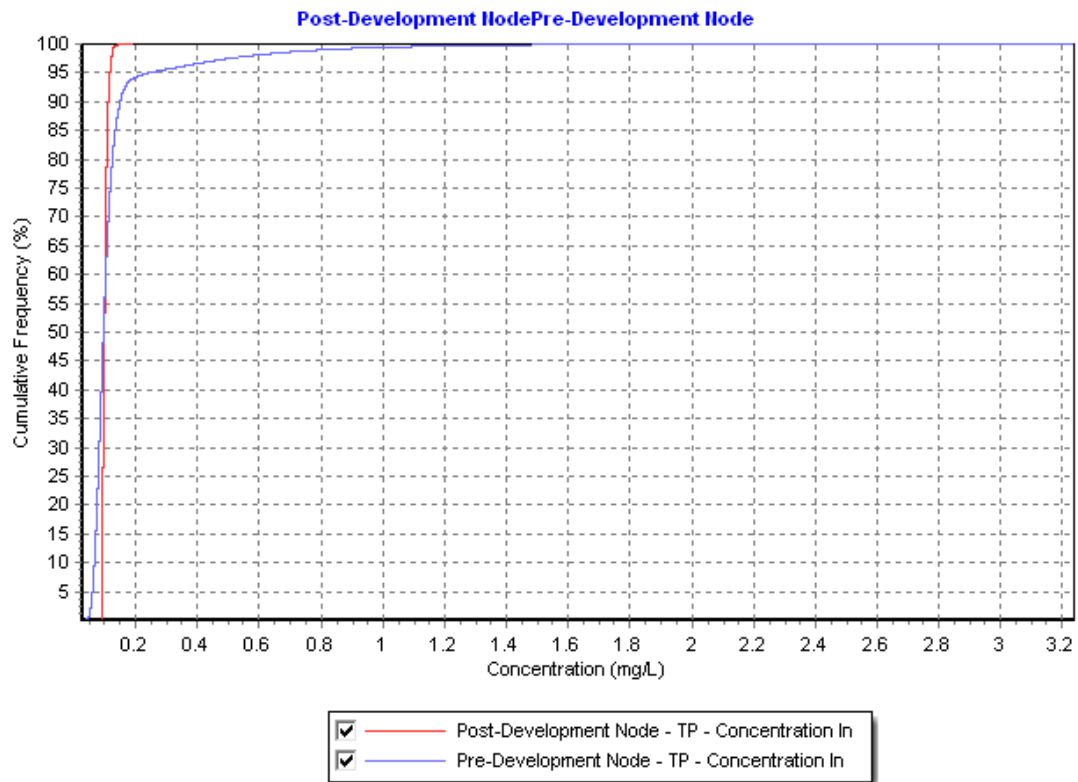


Figure 6.9 Pre-development and Post-development Comparison: Predicted TP Concentrations

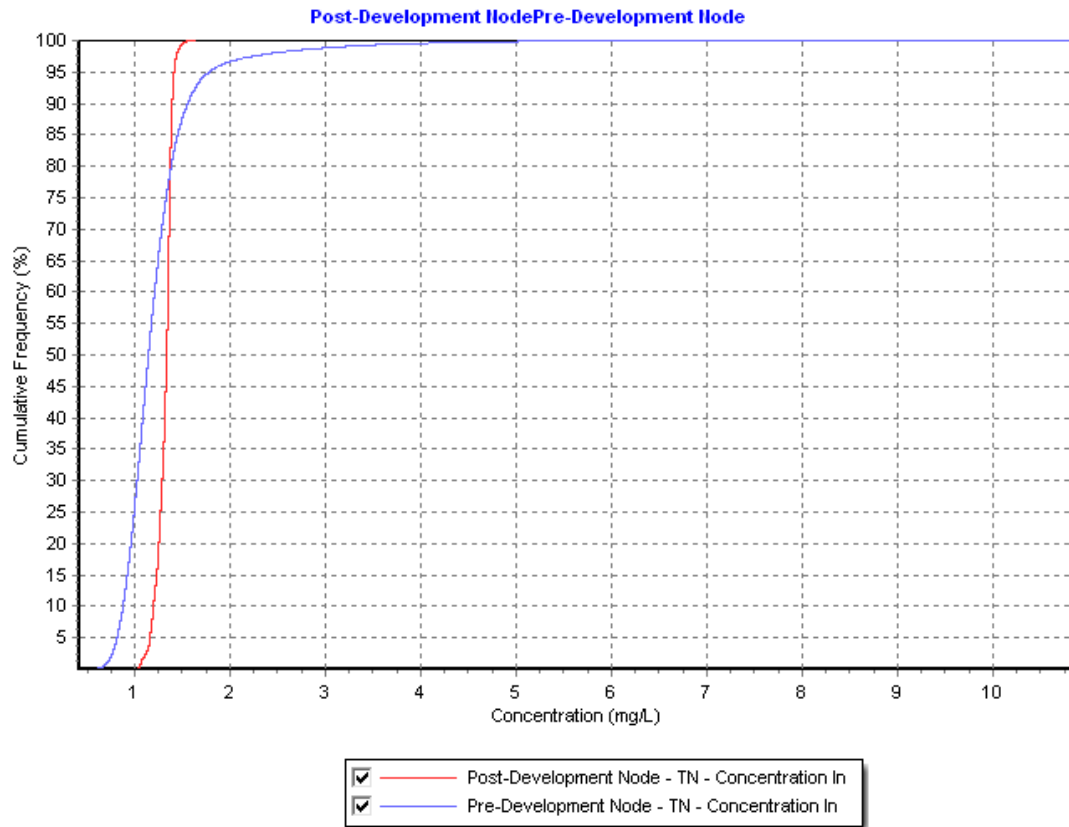


Figure 6.10 Pre-development and Post-development Comparison: Predicted TN Concentrations